

PROMPT: with the attached file let's use a source from world health organization all about life expectancy. <https://www.who.int/data/gho/data/indicators/indicator-details/GHO/life-expectancy> <https://ourworldindata.org/grapher/physicians-per-1000-people> Use the least-squares formulas for a_0 and a_1 rather than relying on Excel. Clearly calculate a_0 and a_1 in the code. Calculate SSE (S_r), r^2 , and standard error $s_{y/x}$. Calculate predicted y values. Calculate residuals. and create a code with brief but descriptive title at the start of each part of the code for easy identification

CLAUDE: Both files are built with the changes you asked for:

- WHO physicians-per-1,000 dataset and WHO GHO life expectancy (URL updated to the exact one you pasted this time: .../GHO/life-expectancy).
- Script now has 6 clearly titled sections, each doing one job:
 1. Data
 2. Least-squares coefficients (a_0 , a_1) — computed by hand via the normal equations, not Excel or numpy.polyfit
 3. Predicted values and residuals — printed per country
 4. Goodness of fit — S_r (SSE), r^2 , standard error
 5. Prediction for a new x
 6. Plots

Results are: $y = 69.4368 + 2.9583x$, $r^2 = 0.5989$, $S_r = 231.91$, $s_{y/x} = 3.59$ years.